WBR Perception

Progress report · 14 October 2026

Known baseline; first camera diagnostics

Pi 5 · D435i · ORB-SLAM3

Accuracy evidence is encouraging.
Physical qualification remains OPEN.



Sensor illustration; no validation claim.

Goal: useful navigation without coupling balance to Linux

PLANNED

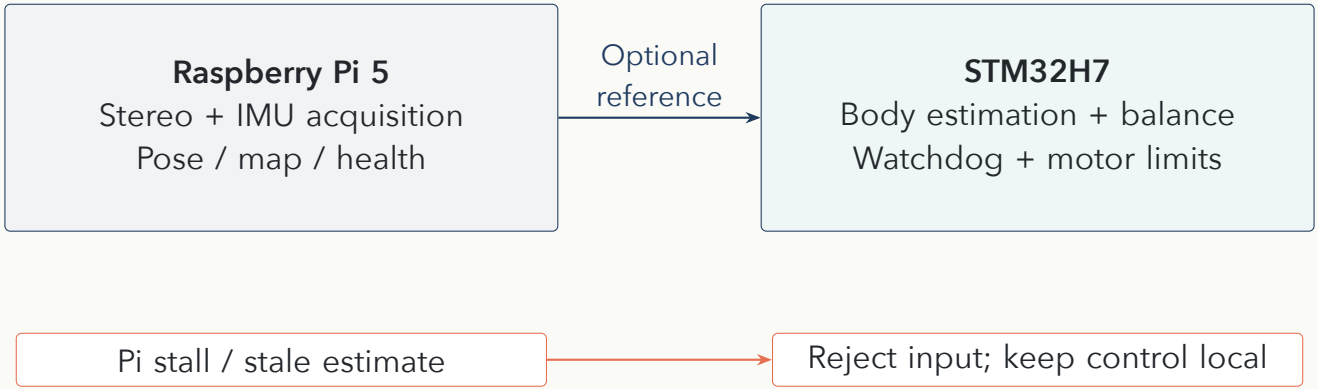
A wheel-biped robot needs scene-relative motion for navigation.

- Pi 5 estimates motion from stereo images and inertial data.
- STM32H7 owns balance, hard-real-time control and motor safety.
- Perception failure must remove optional navigation input safely.

> Qualify perception while preserving the controller ownership boundary.

Ownership: the Pi provides an optional reference

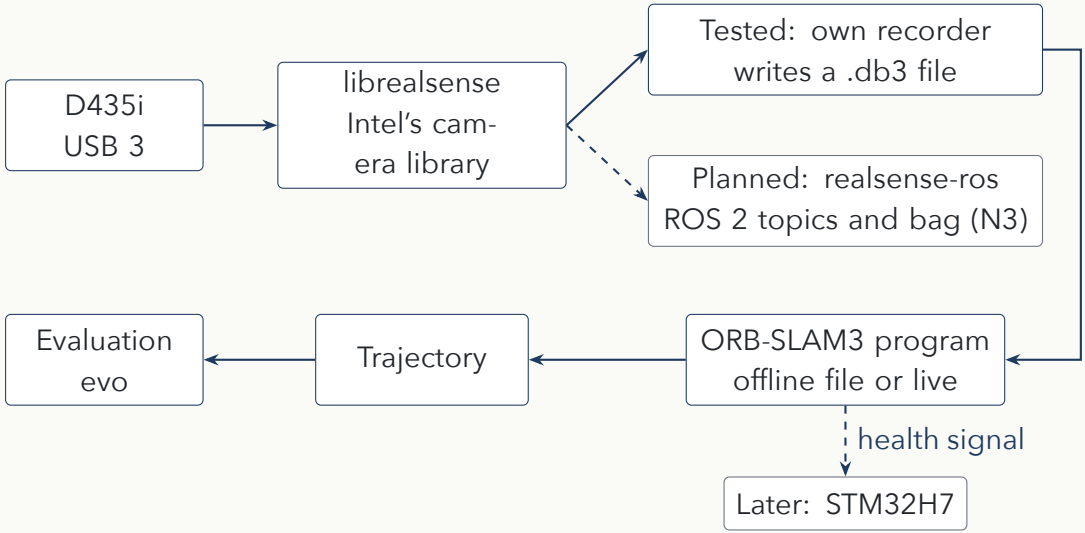
PLANNED / OPEN integration



Receiver health semantics and physical fault tests are still future gates.

Data path: two ways to read the camera

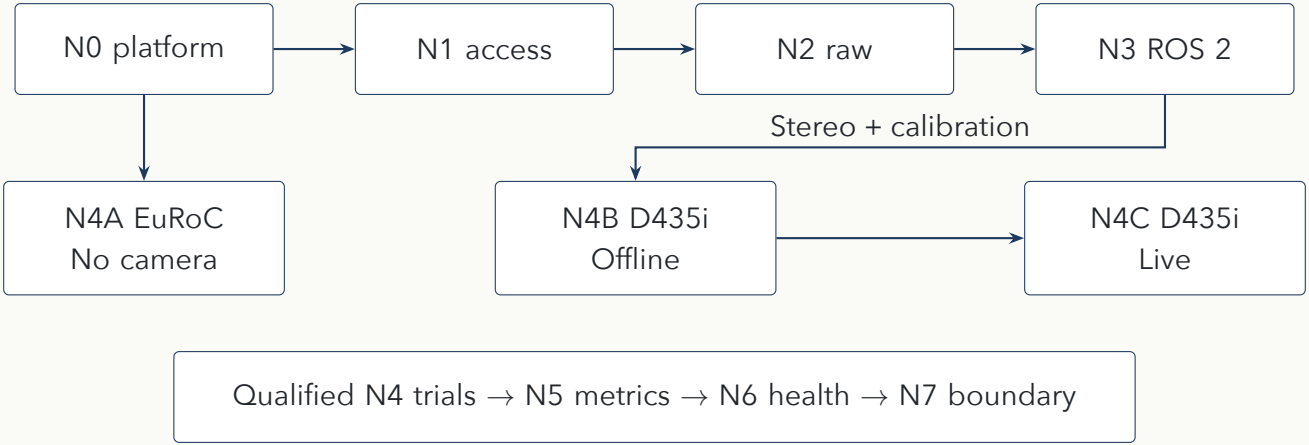
MEASURED recording / OPEN replay / PLANNED ROS 2



Today's two programs use librealsense directly, without ROS 2; the ROS 2 path (N3) is still required.

Method: each gate needs artifacts and acceptance

PLANNED / OPEN



Build success, plausible trajectories and historical receipts do not imply gate PASS.

Gate status: the active unit needs fresh physical evidence

OPEN · as of 14 October 2026

Gate	Current status
N0 platform	OPEN: loaded power unstable
N1 access	OPEN: new unit; prior clock stop
N2 raw	OPEN: new-unit soak missing
N3 ROS 2	OPEN: record/replay unqualified

> Old-unit N1/N2 and calibration receipts do not transfer to the new D435i.

Gate status: measured accuracy; adaptation still OPEN

OPEN / PLANNED · as of 14 October 2026

Gate	Current status
N4A EuRoC	OPEN: three runs; timing invalid
N4B offline	OPEN: first-frame crash; rebuild pending
N4C live	OPEN: built; never camera-run
N5 metrics	PLANNED: full campaign pending
N6 health	PLANNED: semantics/fixtures
N7 boundary	PLANNED: bench review/tests

Since the previous report: a runnable native baseline

MEASURED

Changed	Consequence
New D435i unit	Camera serial `317422076650`; USB serial `327743061057`. Renew physical input and calibration evidence.
Active Pi 5	8 GB board; measured 7.8 GiB RAM; same 32 GB microSD. Same 5 V / 3 A supply: under-voltage still present.
Native ORB-SLAM3	Three build patches; patch 4 applied, rebuild pending. Isolated Pi copy; historical EuRoC example/settings unchanged.

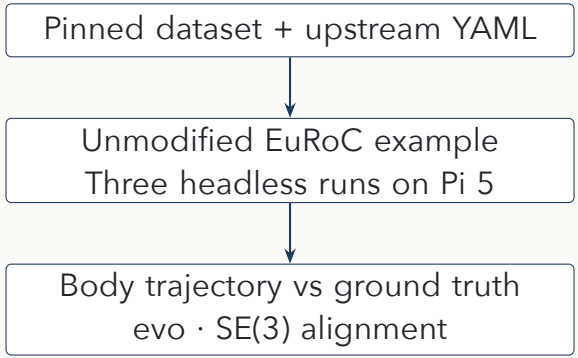
The algorithm path now runs; the hardware change restarts camera qualification.

N4A setup: repeat a dataset with a known answer

MEASURED setup / SOURCE-SUPPORTED dataset

EuRoC MH01

- Public machine-hall drone sequence.
- Stereo 20 Hz + IMU 200 Hz.
- 184 s; 3,682 stereo frames.
- Ground-truth body position.



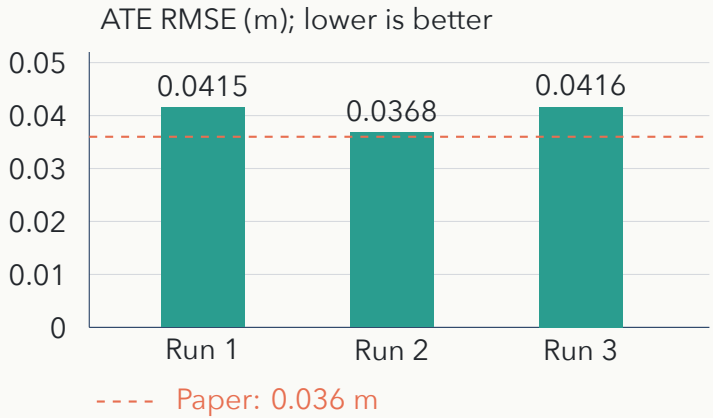
Known input separates native-build behavior from D435i acquisition problems.

N4A result: 0.037–0.042 m translation ATE

MEASURED runs / SOURCE-SUPPORTED paper comparison

Trial	RMSE (m)
Run 1	0.0415
Run 2	0.0368
Run 3	0.0416
Mean	0.040
Paper MH01	0.036

Paper: median of 10 runs.
Our three-run median: 0.0415 m.



All three runs: IMU initialized, one map, no resets; N4A remains OPEN.

N4A supports the native build, with a narrow scope

INFERRED from MEASURED runs / OPEN

Supported on MH01

- ARM64 build produces repeat trajectories.
- Compatibility patches retain similar accuracy.
- Initialization and map continuity are observed.

Still unproven

- Sustained D435i delivery on a moving take.
- Camera–IMU timing and calibration.
- Real-time speed or robot navigation.

Retain the known-data baseline while qualifying each camera-specific input.

D435i recording: short tests work; long take failed

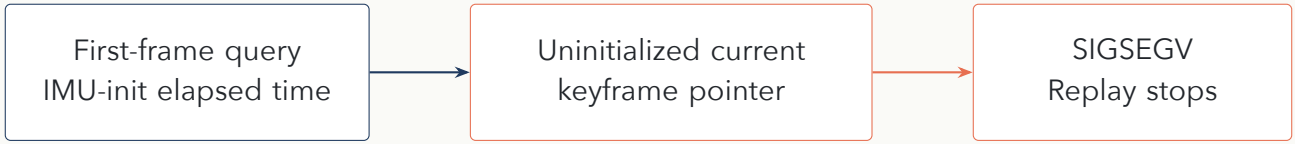
MEASURED tests / INFERRED storage bottleneck / OPEN qualification

Take	Observation
20 s still	Stereo + gyro + accel delivered for 19 s. One ∼0.7 s gap in the final second at shutdown.
100 s hand-moved	Unusable: 0.3–2.9 s gaps after 22 s. Card wrote 16.6 MB/s vs 18.4 MB/s image payload.
15 s into RAM	IR 29.99 Hz; max gap 33 ms; zero dropped frames. Gyro 200.2 Hz; accel 101 Hz; copied with checksum OK.

Record into RAM, then copy: the 100 s RAM take is still unverified.

Offline replay: first-frame crash; patch 4 awaits rebuild

MEASURED crash / SOURCE-SUPPORTED diagnosis / OPEN verification



Patch 4: initialize the pointer to `nullptr`.

Applied in the Pi build copy; power loss interrupted the rebuild.

Resume build, check linkage, then retry offline replay.

EuRoC did not call this query; its three-run result is unchanged.

Blockers: power first, then trustworthy input

MEASURED / ASSUMED / OPEN · 14 October 2026

Risk	Evidence and consequence	What can be done now
Power / timing	4 GB Pi: under-voltage in 49% of samples; timing invalid. 8 GB Pi: under-voltage within 3 min of boot.	**Unresolved.** Use an adequate 5 A supply; retain power logs for each diagnostic run.
Storage	100 s card take dropped frames; RAM test clean for 15 s. USB stick: 1.09 MB/s.	**Demo workaround:** RAM then checksummed copy. Qualified dataset storage still needed.
Calibration	Nominal camera–IMU geometry; IMU noise values copied, not measured.	**OPEN.** Diagnostic room plot has no ground truth; accuracy remains unverified.
RAM	8 GB Pi: 7.8 GiB RAM; RAM recording allocation 3.9 GB.	100 s take ∼1.7 GB fits the budget. Recheck free memory; full take unverified.

15 October: prepare, rebuild, record into RAM

PLANNED diagnostic demo / OPEN exit criteria

	Action	Exit evidence
1	Set up 8 GB Pi	Adequate power; fixed Pi/cables; secure connection. Fresh N1 → N2 → N3 evidence; gates stay OPEN.
2	Resume patch 4 build	Build exits zero; intended rebuilt library is linked. Retain logs and source/binary hashes.
3	Record 100 s into RAM	Smooth early translation + gentle rotation; loop; still ending. Review rates, stamps, gaps, logs and copied-take checksum.

Each step starts only when the previous step's exit evidence exists.

Then: replay, plot, and review before live

PLANNED diagnostic demo / OPEN exit criteria

	Action	Exit evidence
4	Offline replay	Hashed input/settings; initialization; sustained tracking. Review one stable map, resets, queues and nonempty trajectories.
5	Plot and review	Copy evidence to Mac; verify checksums; inspect trajectory. Room take has no ground truth: accuracy unverified.
6	Live, then summary	Proceed only after offline review; same binary/settings. Retain power/kernel logs, counters and finalized outputs.

Nominal-config development runs can inform diagnosis; they cannot qualify VIO.

Summary: baseline retained; camera demo still OPEN

MEASURED progress / OPEN qualification / PLANNED next work

Three EuRoC runs: 0.037–0.042 m ATE.

- Native Pi build is usable on the known-data sequence.
- Short RAM capture works; 100 s capture remains unverified.
- Patch 4 rebuild and successful offline replay are pending.
- Next: RAM take → replay → plot/review → live demo.

> No gate PASS is claimed. STM32H7 retains balance and motor safety.

Further detail: course reference deck and learning deck.

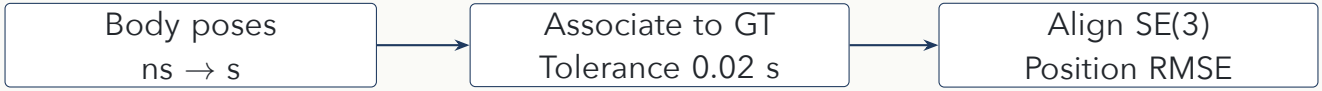
Backup: ATE uses one rigid alignment, without scale

MEASURED evaluation method

Absolute trajectory error: matched body-position differences after alignment.

$$(R^*, t^*) = \arg\min_{R \in SO(3), t} \sum_{i=1}^{n} \|Rp_i + t - q_i\|^2, \qquad \mathrm{ATE}_{\mathrm{RMSE}} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} \|R^* p_i + t^* - q_i\|^2}$$

p_i: estimate; q_i: ground truth; n: matched positions.




```
evo_ape tum gt.tum est.tum -a --t_max_diff 0.02
```

Stereo-inertial has metric scale; Sim(3) would answer a different question.

Backup: three build patches; patch 4 rebuild pending

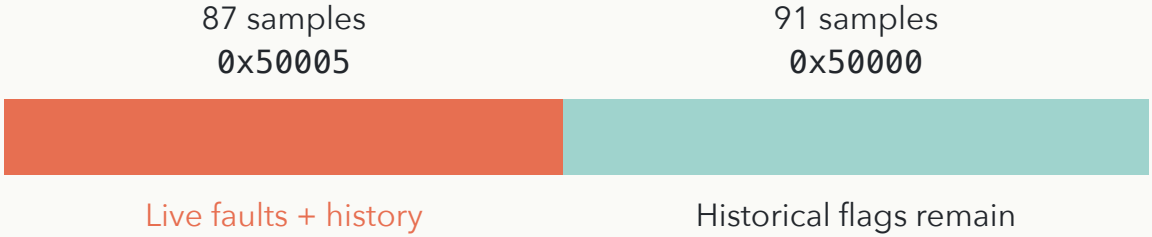
MEASURED build / INFERRED consistency / OPEN patch 4 rebuild

Patch	Reason / limit
C++17; remove `-march=native`	Pangolin 0.9.1 / Ubuntu 24.04 toolchain compatibility. Compile configuration changed; retain source and binary hashes.
Sophus tests OFF	GCC 13 aarch64 array-bounds warning fails their `-Werror`. ORB-SLAM3 uses Sophus headers; disabled tests are not validated.
`mnFullBAIdx++` → `= true`	C++17 forbids incrementing bool. Preserves the prior C++11 bool-increment effect.
Current keyframe → `nullptr`	Patch 4 avoids the first-frame uninitialized-pointer query. Applied in Pi copy; rebuild and replay unverified.

MH01 accuracy is evidence for this build; it does not validate every code path.

Backup: live power faults in 87 of 178 samples

MEASURED samples / CALCULATED 48.9%



5-second sampling during the EuRoC runs.

Maximum observed temperature: about 63 °C.

Pi UTC labels were about 80 minutes behind; NTP corrected later.

Historical-only flags are not a clean N0 result; timing must be repeated.

Backup: stereo payload already consumes the old write margin

CALCULATED payload / MEASURED tests / OPEN storage

$$640 \times 480 \times 1 \times 2 \times 30 = 18{,}432{,}000\ \mathrm{B/s} \approx 18.4\ \mathrm{MB/s}$$

Quantity	Implication
100 s stereo payload	1.84 GB before overhead; failed take wrote 1.66 GB.
100 s card take: 16.6 MB/s	Repeated gaps; microSD blocking strongly inferred.
RAM: 3.9 GB allocation	15 s test clean; 100 s ∼1.7 GB budget remains unverified.
USB stick: 1.09 MB/s	Too slow for recording; qualified dataset storage still needed.

Qualify storage under load and copy takes to the Mac after each session.

References

C. Campos, R. Elvira, J. J. Gómez Rodríguez, J. M. M. Montiel, J. D. Tardós. ORB-SLAM3: An accurate open-source library for visual, visual–inertial and multi-map SLAM. *IEEE Transactions on Robotics*, 37(6), 2021. arXiv:2007.11898

M. Burri et al. The EuRoC micro aerial vehicle datasets. *International Journal of Robotics Research*, 35(10), 2016.

Project evidence (experiment reports, logs and trajectories) is listed per slide in the speaker notes.